Statistics for Scientists

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Statistics for Scientists (A11491)

Short Title:	Statistics for Scientists		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
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Credits	5	**Level:**	Introductory

Description of Module / Aims

This module introduces the student to some fundamental statistical concepts, to probability and sampling mechanisms as well as basic methods in descriptive and inferential statistics and regression.

Programmes

SCIE-0015	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 3 / M
SCIE-0015	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	2 / 3 / M
SCIE-0015	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	2 / 3 / M
SCIE-0015	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 3 / M
SCIE-0015	BSc in Food Science with Business (WD_SFDSC_D)	2 / 3 / M
SCIE-0015	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	2 / 3 / M
SCIE-0015	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 3 / M

Indicative Content

- Introduction to statistics: types of variables (scale, nominal, ordinal); predictor versus response and observational versus experimental variables; descriptive and inferential statistics; population and sample; introduction to probability
- Sampling: probability versus non-probability; simple random; stratified; cluster; systematic; convenience etc., randomising the run order of experiments.
- Descriptive statistics: statistics measuring centre (mean, median, mode) and spread (standard deviation, quartiles); charts for categorical data (bar charts, pie charts) and scale data (histogram, stem and leaf plot, boxplot); analysis of outliers
- Normal distribution: calculating probabilities and quartiles; verifying normality-probability plots; other distributions
- Statistical Inference: null and alternative hypothesis; p-values; confidence intervals for means; t-tests; two sample paired and independent and ratio of variances; interpreting ANOVA table
- Linear regression: parameter estimation using graphical and analytical methods; correlation coefficient; prediction; interpolation and extrapolation
- Using a suitable software application such as Minitab or MS Excel: selecting appropriate methods correctly; interpreting output meaningfully

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe and summarize the nomenclature and classifications of introductory statistics.
2. Interpret appropriate descriptive statistics constructs and graphs to summarize variables of different types.
3. Discuss methods of sampling.
4. Compute using a calculator and tables, tests of statistical significance and interpret the analysis.
5. Apply and appraise regression models.
6. Determine the goodness of fit of the normal distribution model to sample data.
7. Calculate elementary statistical analysis and interpret the output using industrial standard statistical software (e.g. Minitab).

Learning and Teaching Methods

- Lectures.
- In class exercises.
- Discussion.
- Self-directed learning is emphasized. Students are supported with notes, online material and feedback.
- Practicals using statistical software supports the learning experience.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5,6
Continuous Assessment	30%	
In-Class Assessment	30%	7

Assessment Criteria

<40%: Very limited knowledge and understanding of descriptive and inferential statistics and regression.

40%-49%: Demonstrate a limited knowledge of descriptive and inferential statistics and regression..

50%-59%: Demonstrate satisfactory general knowledge of the main issues within descriptive and inferential statistics and regression.

60%-69%: Demonstrate sound knowledge of descriptive and inferential statistics and regression. Show ability to analyse and logically argue in an effective and mature style.

70%-100%: Demonstrate authoritative handling of complex material and provide well focused analysis and convincing arguments on statistics and regression.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- "Looking for Info on Statistics." http://www.statsoft.com/textbook/
- Dukkipati, R.V. _Probability and Statistics: For Scientists and Engineers (__ebook__)_. US: New Academic Science Limited, 2013.
- Reilly, J. _Applied Statistics (__ebook__)_. Ireland: Statistical Solutions, 2015.
- Reilly, J. _Understanding Statistics_. Dublin: Folens, 1997.

Requested Resources

- Room Type: Computer Lab